

Category 1

Python Fundamentals + Data Cleaning

Skills expected:

variables, input/output, conditions, `if/elif/else`, loops, lists/dictionaries, functions, string handling, missing values, duplicates, inconsistent data, basic Pandas/data cleaning.

NOTE : Solve Both Problems

Problem 1 :

Smart Attendance System

Create a simple attendance analysis program.

Given attendance records such as:

Ali: Present
Sara: Absent
Ahmed: Present
...

Requirements:

- Count present and absent students.
- Calculate attendance percentage.
- Identify students below 75%.
- Display attendance status:
 - $\geq 90\%$ → Excellent
 - 75–89% → Good
 - $< 75\%$ → Warning

Problem 2:

Inventory Management System

Create a small shop inventory program.

Each product should have:

- Product name
- Price
- Quantity

Requirements:

- Display all products.
- Add a product.
- Update quantity.
- Calculate total inventory value.

Criteria	Marks
Problem understanding	15
Python implementation	25
Control flow/loops	15
Data structures	15
Data cleaning	20
Output/presentation	10
Total	100

Category 2

Machine Learning

Skills expected:

train/test split, feature selection, preprocessing, Linear Regression, Logistic Regression, Decision Tree, Random forest

Problem

Customer Churn Prediction

A bank is experiencing customer loss. Some customers are closing their accounts or leaving the bank for competitors.

The bank wants to develop an **AI-powered Customer Churn Prediction System** that can identify customers who are likely to leave.

[Dataset Of customers](#)

Hackathon Task:

Your task is to clean the data, analyze customer behavior, train machine-learning models, and develop a system that predicts whether a customer is likely to leave the bank.

Your Mission

- Develop a machine-learning solution that answers "**Which customers are likely to leave the bank?**"
- Your solution should take customer information as input and predict whether that customer is likely to churn.

Target: Churn = Yes/No

Phase 1

Perform an exploratory analysis of the dataset.

Determine:

- Number of rows and columns
- Numerical and categorical features
- Missing values
- Duplicate records
- Basic statistics

Create at least **3 meaningful visualizations**.

Customer age distribution

Credit-score distribution

Churn by geography

Phase 2

Data Cleaning

The dataset contains data-quality problems.

You must:

1. Detect missing values.
2. Decide how to handle missing numerical values.
3. Handle missing categorical values.
4. Identify inconsistent text values.
5. Remove unnecessary spaces from categorical data.
6. Check for duplicate records.
7. Identify columns that should **not** be used as predictive features.

Phase 3

Feature Engineering

Students should prepare the data for machine learning.

Required:

- Encode categorical variables.
- Select relevant features.
- Separate features X and target y .
- Split data into training and testing sets.

Phase 4

Machine Learning

You must train **at least three models**:

Logistic Regression Decision Tree Random Forest

Phase 5

Model Evaluation

Do not evaluate the models using accuracy alone.

Calculate:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix

Category 2 100 Marks

Criteria	Marks
Problem understanding	10
Data preprocessing	20
Model implementation	20
Model evaluation	15
Model comparison	15
Total	80

For **Category 2**, we don't judge students purely on accuracy. A student who gets 95% accuracy by using an inappropriate split or causing data leakage should not beat someone who builds a properly validated 90% model.